Majorana Modes in the Machine

Week 9: Gate Errors via Qiskit – Verification and Topological Protection

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Advanced Seminar – AI-Augmented Theoretical Physics

June 20, 2026

From Week 8 to Week 9

Week 8 (frozen-state test)

- Took the validated even-parity ground state $\rho_0(\mu)$.
- Applied a *hand-rolled* two-qubit depolarizing channel once per neighbor pair.
- Added symmetric readout and shots.

Week 9 goals

- Integrate gate errors at the *circuit* level via Qiskit's `NoiseModel`.
- **Verify** the implementation is correct.
- Ask whether the frozen-state model is faithful to hardware.
- Show **where topology helps**.

Same physics throughout: same Hamiltonian and edge string $O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ as Block 3; only the noise treatment changes.

Gate Errors as Quantum Channels

A noisy gate is the ideal unitary followed by a CPTP error channel:

$$\tilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \qquad \mathcal{E}_g(\rho) = \sum_k K_k \, \rho \, K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = I. \tag{1}$$

A full circuit composes these gate-by-gate:

$$\rho_{\mathrm{out}} = \left(\mathcal{E}_{g_M} \circ \mathcal{U}_{g_M}\right) \circ \cdots \circ \left(\mathcal{E}_{g_1} \circ \mathcal{U}_{g_1}\right)(\rho_{\mathrm{in}}). \tag{2}$$

Key consequence

Noise enters *through the gates*: the number of error channels grows with the gate count and circuit depth. This is exactly what a single channel on a frozen state cannot capture.

Two Conventions, One Channel

Week 8 (random non-identity Pauli), per two-qubit pair:

$$\mathcal{D}_q(\rho) = (1-q)\,\rho + \frac{q}{15}\sum_{P\neq II} P\rho P. \tag{3}$$

Qiskit `depolarizing_error(p,2)` **(replace with maximally mixed):**

$$\mathcal{E}_p^{\mathrm{Qiskit}}(\rho) = (1-p)\,\rho + p\,\frac{\operatorname{Tr}\rho}{2^n}\,I = \left(1-\tfrac{15p}{16}\right)\rho + \frac{p}{16}\sum_{P\neq II} P\rho P. \tag{4}$$

Matching the non-identity weight $\frac{p}{16} = \frac{q}{15}$ gives a single map:

$$p_{\mathrm{cx}} = \frac{2^{2n}}{2^{2n}-1}\,q \;\Rightarrow\; \tfrac{16}{15}\,q \text{ (two-qubit)}, \quad \tfrac{4}{3}\,q \text{ (one-qubit)}.$$

Integrating and Verifying in Qiskit

```
from qiskit_aer.noise import NoiseModel, depolarizing_error

nm = NoiseModel()
# Week 8 channel == Qiskit error under the 16/15 map
err = depolarizing_error(16/15 * p2q, 2)
nm.add_all_qubit_quantum_error(err, ['cx'])
```

Verification (`block4.py --plots 2`, on a random two-qubit ρ):

- mapped parameter $\frac{16}{15}q$: $\max|\Delta\rho| = 1.4 \times 10^{-16}$ (machine precision)
- naive guess $p = q$: $\max|\Delta\rho| = 7.9 \times 10^{-4}$ (wrong)

The $\frac{16}{15}$ factor is physically required

Any quantitative comparison between the Week 8 plot and a Qiskit `NoiseModel` must apply it.

Is the Frozen-State Model the Real One?

1. It is depth-blind.

- A linear `EfficientSU2` ansatz with r reps has $r(L-1)$ CNOTs.
- Real noise insertions grow with depth.
- Week 8 applies only $L-1$ channels, *once*, regardless of the circuit.

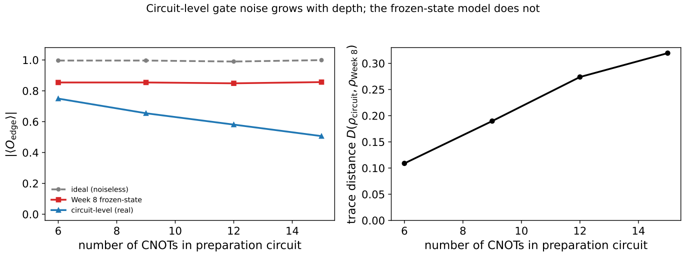
2. Wrong state, wrong order.

- Real errors occur *between* gates and are scrambled by later unitaries.
- Week 8 applies them once, at the end, to the perfect state.
- $\mathcal{E} \circ \mathcal{U} \neq \mathcal{U} \circ \mathcal{E}$ – ordering matters.

Claim

The Week 8 frozen-state result is an *optimistic bound*, not a faithful model of hardware gate noise.

Demonstration: Noise Grows with Depth



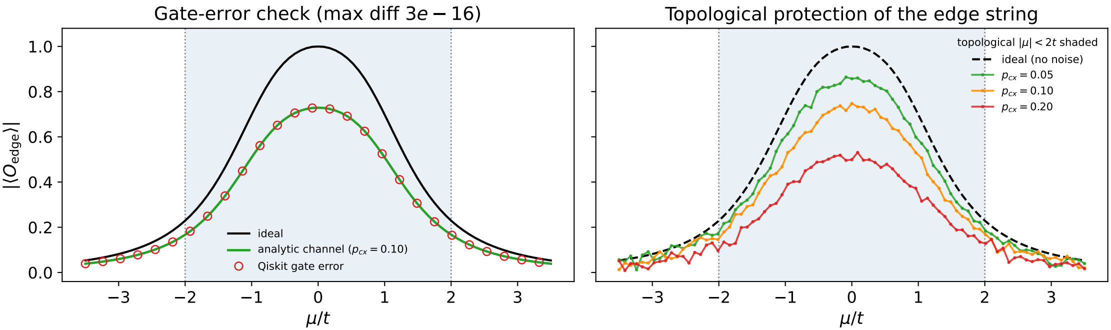
Same θ, matched per-channel strength ($\mu = 0$):

CNOTs	circ.	W8	D
6	0.75	0.85	0.11
9	0.66	0.85	0.19
12	0.58	0.85	0.27
15	0.51	0.86	0.32

Week 8 stays *flat*; the real circuit *decays*. At depth 5 it over-estimates the signal by $\sim 70\%$.

Verification Across the Sweep + Topological Protection

Verified gate error and the protected topological plateau (frozen state, shots)



Left: analytic channel (line) vs Qiskit gate error (markers) agree to 3×10^{-16} over all μ. **Right:** verified noise at three strengths; the topological plateau ($|\mu| < 2t$) is suppressed but stays distinguishable from the trivial floor, most robustly at the $\mu = 0$ sweet spot.

What "Topology Protects" Means Here

- The edge string measures the Majorana edge mode, which exists only for $|\mu| < 2t$ and is protected there by the bulk gap.
- Under noise the topological signature is reduced in amplitude but not in *location*: the phase distinction survives.
- The transitions $|\mu| = 2t$ (gap closing) are the fragile points.

Careful statement

Decoherence is *not* suppressed by topology – without error correction, gate noise always lowers contrast. What the gap protects is the *existence and localization* of the edge mode, so its signature remains measurable, most robustly deep in the topological phase.

What This Enables Next

Now that gate errors are integrated and verified

- **Noisy VQE optimization**: optimize with a `NoiseModel`-backed estimator, so every cost evaluation is corrupted.
- **Backend-calibrated noise**: `NoiseModel.from_backend(...)` for real device T_1/T_2 and gate errors.
- **Readout/SPAM layer**: fold the asymmetric assignment matrix into the same circuit-level model.

Full derivations and code: `notes/week9_note.tex`, `src/block4.py` (plots 2–3).